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Question 1

Correct

Mark 3.00 out of 3.00

Principles of Learning Theory

Which type of learning process is involved in each of the following situations?

The fear that a child with a spider phobia shows towards insects in general.	Stimulus generalisation ✔
Humans usually react with a fear response to snakes.	Stimulus preparedness ✔
Cars have the habit of flashing a seat belt warning sign until you fasten your seat belt.	Negative reinforcement ✔

Your answer is correct.

Stimulus preparedness (Seligman) explains why snake and spider phobia are commoner than shoe phobia or watch phobia. In evolutionary terms, the stimuli that were threatening to hunter-gatherer man has been hard-wired in our system to respond immediately - and phobia develops more readily for such 'prepared stimuli'.

Pavlovs' dog experiments were conducted using human subjects by Watson & Rayner and paved the way for founding the school of behaviourism. Watson produced 'phobia' in an infant called Little Albert. By exposing him to loud, frightening noise whenever he was shown a white rat, eventually Albert became fearful of the white rat, even when he heard no loud noise. A similar fear response was seen when any furry white object was shown to Albert. This 'spread' of associative learning from one stimulus to other is called **stimulus generalisation**.

Reinforcement can be positive or negative according to the way reinforcers are handled i.e. if something is given or taken away. If a particular behaviour increases by providing a reinforcer, then this is positive reinforcement. If a particular behaviour increases by removal of an aversive stimulus then this is negative reinforcement. Similarly punishment can also be positive (smack) or negative (take the chocolate away) though it is often used in positive sense.

The correct answer is: The fear that a child with a spider phobia shows towards insects in general. → Stimulus generalisation, Humans usually react with a fear response to snakes. → Stimulus preparedness, Cars have the habit of flashing a seat belt warning sign until you

fasten your seat belt. → Negative reinforcement

Question 2

Correct

Mark 3.00 out of 3.00

Adverse effects

For each of the following drugs, identify the most important side effect associated with its therapeutic use

Lithium	Hypothyroidism ✔
Aripiprazole	Nausea, vomiting ✔
Topiramate	Word-finding difficulty ✔

Your answer is correct.

Lithium: can cause a variety of thyroid problems - the most common being a **benign hypothyroid state**. 5% patients may develop goiter, and overt hyperthyroidism is also reported in some cases. Thyroid deficiency is common in those with high risk for preexisting antithyroid antibodies such as middle-aged women. The risk is 3-4:1 in women and is high in first two years of treatment. Rapid-cycling patients are at higher risk. High TSH is seen in nearly 1/3rd of chronic lithium-treated patients - without clinical hypothyroidism. In resistant depression or non-responsive cyclers with bipolar it may be useful treating this subclinical state with thyroxine.

Aripiprazole: a D2 receptor partial agonist and therefore, more likely to induce side effects like **nausea and vomiting.**

Topiramate: Several studies have documented language impairment due to the use of topiramate. Word-finding difficulties are noted in 7.2% of epileptic patients taking topiramate, especially those with a positive family history of epilepsy, or simple partial seizures and left temporal seizures.

The correct answer is: Lithium → Hypothyroidism, Aripiprazole → Nausea, vomiting, Topiramate → Word-finding difficulty

Question 3

Correct

Mark 6.00 out of 6.00

MoK Neurosciences CXEMI009

Neuropathology

For the following clinical conditions, identify the most common neuropathological changes in the brain.

1. A 70-year-old gentleman with history of parkinsonism, confusion and visual hallucinations (Choose TWO)

Sensitivity to antipsychotics ✔

Intraneuronal synuclein deposits in cortex and substantia nigra ✔

2. An 85-year-old lady with progressive memory loss for recent events and dysphasia, which has become more generalized. She also has a 20-years history of alcohol misuse (Choose TWO) Neurofibrillary tangles and amyloid plaques in neocortex ✔

Presence of e-4 allele of apo-E4 gene ✔

3. A 57-year-old gentleman presenting with altered behaviour, apathy, disinhibition and word finding difficulties. Mother and aunt died from a similar condition. (Choose TWO) Atrophy of frontal poles ✔

Mutation in Tau gene ✔

Explanation

Case 1: This is a case of **Lewy body dementia** characterised by **neuroleptic sensitivity** and the presence of **Lewy bodies** on pathological examination. Lewy bodies are intraneuronal deposits.

Case 2: This is an example of **Alzheimer's disease** characterised by **amyloid plaques, neurofibrillary tangles** and **apo-E4 gene**.

Case 3: This case describes **frontotemporal dementia** characterised by **frontal and temporal lobe atrophy** and **mutations in the tau gene**.

The correct answer is:

A 70-year-old gentleman with history of parkinsonism, confusion and visual hallucinations (Choose TWO) - Sensitivity to antipsychotics, Intraneuronal synuclein deposits in cortex and substantia nigra,

An 85-year-old lady with progressive memory loss for recent events and dysphasia, which has become more generalized. She also has a 20-years history of alcohol misuse (Choose TWO) - Presence of e-4 allele, Neurofibrillary tangles and amyloid plaques in neocortex,

A 57-year-old gentleman presenting with altered behaviour, apathy, disinhibition and word finding difficulties. His mother and aunt died from a similar condition. (Choose TWO) - Atrophy of frontal poles, Mutation in Tau gene

Question 4

Correct

Mark 3.00 out of 3.00

Selectivity of Tricyclics (TCAs)

Identify the most suitable drug, for the descriptions given below:

This drug is the most serotonergic reuptake-selective of all tricyclic antidepressants.	Clomipramine ✔
This drug is the most noradrenergic-selective of all tricyclic antidepressants.	Desipramine ✔
This drug has the most antihistaminic activity.	Doxepin ✔

Your answer is correct.

Clomipramine more serotonergic reuptake selective; **desipramine** is the most noradrenergic selective of TCAs. Amoxapine, nortriptyline, desipramine, and maprotiline (secondary amines) have the least anticholinergic activity; **doxepin** has the most antihistaminic activity.

The correct answer is: This drug is the most serotonergic reuptake-selective of all tricyclic antidepressants. → Clomipramine, This drug is the most noradrenergic-selective of all tricyclic antidepressants. → Desipramine, This drug has the most antihistaminic activity. → Doxepin

Question 5

Partially correct

Mark 2.00 out of 3.00

MoK Pharmacology CXEMI009

ECG changes

Psychotropic use is often associated with ECG changes, necessitating a good working knowledge of ECG findings for a psychiatric trainee. For each of the following clinical conditions, identify the possible ECG changes.

1. A 25-year-old patient with chronic schizophrenia was started on clozapine two weeks ago. He developed flu-like symptoms and chest pain. Pulse rate is 120 per minute, but it is regular. (Choose TWO)

Diffuse ST elevation with saddle formation ✔

P-R interval of 100 ms, P waves present before every QRS ✖

2. A 79-year-old woman with a history of falls and blackouts has an MMSE score of 14/30. She was recently started on medications for memory problems. He pulse is reduced now at a rate of 45/minute. (Choose ONE)

P-R interval of 280 ms, P waves present before every QRS ✔

Explanation

Case 1: This is a case of **clozapine-induced myocarditis**. It generally occurs within 6-8 weeks of starting clozapine treatment. **Saddle elevation of ST waves** is due to pericardial involvement. **Diffuse T wave inversion** can also be seen. The normal PR interval of 120 to 200ms (3 - 5 small squares) is altered resulting in an abnormal **prolongation of PR** along with tachycardia. In some cases, complete heart block may ensue.

The correct answer is: P-R interval of 280 ms and P waves present before every QRS, Diffuse ST elevation with saddle formation

Case 2: This is a patient who is started on **acetylcholinesterase inhibitors** for dementia. This group of drugs often causes **prolongation of PR interval** on ECG along with **sinus bradycardia.**

The correct answer is:

P-R interval of 280 ms, P waves present before every QRS

Question 6

Correct

Mark 6.00 out of 6.00

MoK Neurosciences CXEMI003

Localizing signs

Choose two symptoms for each of the following conditions:

1. A malnourished alcoholic. (TWO) Gait Ataxia ✔ Bilateral 6th nerve palsy ✔
2. A right-handed person with right parietal lobe lesion. (TWO) Constructional Apraxia ✔ Anosognosia ✔
3. A frontal meningioma. (TWO) Anosmia ✔ Ipsilateral optic atrophy ✔

Explanation

Case 1: This patient is most likely to have **Wernicke's encephalopathy**. The classic triad includes **ophthalmoplegia**, **ataxia** and **confusion**. The most common ophthalmoplegia here is a bilateral 6th nerve (not 4th nerve) lesion. The 4th nerve may be affected in few cases.

Case 2: The **non-dominant parietal lobe** has a significant role in **constructional praxis** and also in **awareness of illness states**.

Case 3: **Frontal meningiomas** are associated with **anosmia** (often present due to the common olfactory groove distribution of tumours causing pressure) and **ipsilateral optic atrophy**, on the side of the tumour. Anosmia with ipsilateral optic atrophy and contralateral papilloedema is also known as Kennedy-Foster syndrome.

The correct answer is:

A malnourished alcoholic. (TWO) - Gait Ataxia, Bilateral 6th nerve palsy

A right-handed person with right parietal lobe lesion. (TWO) - Anosognosia, Constructional Apraxia

A frontal meningioma. (TWO) - Ipsilateral optic atrophy, Anosmia

Question 7

Correct

Mark 8.00 out of 8.00

QTc prolongation

Mr Barnes is a 78-year-old man with a diagnosis of schizophrenia and moderate depression. He has been trialled on a number of medications, some of them with only limited success. You are reviewing his notes and deciding on the next possible pharmacotherapeutic options. You are aware he has a family history of cardiac dysfunction, with ischaemic heart disease on the maternal side, and cardiac arrhythmia on the paternal side. Choose from the list of medications given, that has:

1. No effect on the QTc at therapeutic doses (choose 2 options) Aripiprazole ✔ Mirtazapine ✔
2. Low effect on the QTc at therapeutic doses (choose 2 options) Venlafaxine ✔ Risperidone ✔
3. Moderate effect on the QTc at therapeutic doses (choose 2 options) Citalopram ✔ Quetiapine ✔
4. High effect on the QTc interval at therapeutic doses (choose 2 options) IV Haloperidol ✔ Sertindole ✔

1) **No effect** on the QTc at therapeutic doses- Drugs include **Aripiprazole** and **Mirtazapine**

2) **Low effect** on the QTc at therapeutic doses- Drugs include **Venlafaxine** and **Risperidone**.

3) **Moderate effect** on the QTc at therapeutic doses- Drugs include **Citalopram** and **Quetiapine**.

4) **High effect** on the QTc at therapeutic doses- Drugs include **Haloperidol**, **Sertindole** and **any intravenous antipsychotic.**

(Ref: The Maudsley Prescribing guidelines- 11th edition- page 117)

Question 8

Correct

Mark 3.00 out of 3.00

Features of defence mechanisms

For each of the characteristic features described below chose the most appropriate defence mechanism suitable from the above list

Moving consciously available material to the unconscious in a way that the material cannot be recalled at will when required	Repression ✔
Moving consciously available material to the unconscious in a way that the material can be recalled at will when required	Suppression ✔
Expressing feelings without causing unpleasantness or distress to others	Humour ✔

Your answer is correct.

Explanation:

In repression, recalling at will is absent. Repression can form the basis of dissociative / conversion disorders. By contrast, suppression is a mature defense that involves conscious postponement of an issue - recalling at will is possible in suppression. Focusing on funny aspects of a painful situation is called humour. It is a mature defence strategy.

In projection, intolerable feelings, impulses or thoughts are falsely attributed to other people. Passive aggression can manifest itself as repeated postponements, rigidity in task performance, resentment, sullenness, or repeated failure to take up the responsibility to complete tasks.

The correct answer is: Moving consciously available material to the unconscious in a way that the material cannot be recalled at will when required → Repression, Moving consciously available material to the unconscious in a way that the material can be recalled at will when

required → Suppression, Expressing feelings without causing unpleasantness or distress to others → Humour

Question 9

Correct

Mark 3.00 out of 3.00

DSM Axes

Identify the descriptions from the given list that are best suited to each of the following DSM-IV axes.

Axis-2	Schizoid traits	✔
Axis-3	Hypertension	✔
Axis-5	Global assessment of functioning=60	✔

Your answer is correct.

The multi-axial system of DSM-IV used five axes. Though these axes are now not employed in DSM-5, the multiaxial concept is still being used by researchers and health service analysts.

Axis I - Clinical Disorders (e.g. anorexia, alcohol intoxication, psychosis due to medical condition)

Axis II - Personality Disorders/Mental Retardation (e.g. moderate mental retardation, anxious personality traits)

Axis III - General Medical Conditions (e.g. epilepsy, hypertension)

Axis IV - Psychosocial and Environmental Problems (e.g. unemployment)

Axis V - Global Assessment of Functioning (e.g. GAF score of 60)

The correct answer is: Axis-2 → Schizoid traits, Axis-3 → Hypertension, Axis-5 → Global assessment of functioning=60

Question 10

Correct

Mark 3.00 out of 3.00

Clinical neurological signs

Identify the neurological sign seen in each of the following situations:

A 60-year-old man develops alcohol related hepatic failure, which results in anoxic brain damage.	Flapping tremor ✔
A 35-year-old man is a known binge drinker with significant malnourishment. He present with acute onset confusion and disorientation.	Ataxia ✔
An 38-year-old woman developed opthalmoplegia. A T1-weighted MRI reveals multiple hypointense lesions in the corpus callosum.	Lhermitte's sign ✔

Your answer is correct.

A flapping tremor is seen in cases of respiratory, hepatic or cardiac failure with encephalopathy.

Ataxia and ophthalmoplegia with nystagmus on a background of impaired consciousness are characteristic of Wernicke's encephalopathy.

Lhermitte's sign is characterised by a sudden unpleasant (but not painful) electric shock-like sensation starting in the neck and radiating along the spine towards distal limbs. It is seen in patients with multiple sclerosis.

The correct answer is: A 60-year-old man develops alcohol related hepatic failure, which results in anoxic brain damage. → Flapping tremor, A 35-year-old man is a known binge drinker with significant malnourishment. He present with acute onset confusion and disorientation. → Ataxia, An 38-year-old woman developed opthalmoplegia. A T1-weighted MRI reveals multiple hypointense lesions in the corpus callosum. → Lhermitte's sign

Question 11

Correct

Mark 3.00 out of 3.00

Alcohol-induced physical symptoms

Choose the symptoms which would best support each diagnosis in an alcoholic, malnourished middle-aged man who becomes confused two days after emergency surgery for a femoral shaft fracture.

Bacterial septicaemia	Neck stiffness	✔
Wernicke's syndrome	Ophthalmoplegia	✔
Acute hepatic failure	Jaundice	✔

Your answer is correct.

Please note that this patient has recently had an emergency surgery of the femur. Don't try to do Kernig's test on him; also gait ataxia may not be demonstrable. Petechial rashes and neck stiffness may be observed when septicaemia is present.

Wernicke's syndrome is an acute condition characterised by the well-known triad of ophthalmoplegia, ataxia and global confusional state. The classic triad is only present in 10% of patients, and 80% of patients are not diagnosed before postmortem. The eye signs are often bilateral though need not be always symmetrical.

Jaundice and flapping tremor (asterixis) are features of hepatic failure. Asterixis can be elicited by having the patient extend both arms with the wrists dorsiflexed and palms facing forward, and eyes closed. Brief jerky downward movements of the wrist are considered a positive sign. Asterixis is commonly seen with metabolic encephalopathies. Note: pronator drift is elicited by having the patient extend both arms with the wrists supinated and palms facing upwards and eyes closed - slow unequal drift towards pronation indicates hemiparetic weakness. Elevated JVP is not seen in hepatic failure that is due to non-cardiac causes.

The correct answer is: Bacterial septicaemia → Neck stiffness, Wernicke's syndrome → Ophthalmoplegia, Acute hepatic failure → Jaundice

Question 12

Correct

Mark 3.00 out of 3.00

Dementia
Choose salient features for each item below from the list shown:

A salient features of Lewy Body dementia	Dementia with prominent but short lucid intervals ✔
A salient feature of CJD	Rapidly progressing dementia ✔
A salient feature of dementia with Parkinson's disease	Emotional blunting ✔

Your answer is correct.

Dementia with Lewy Bodies is characterised by spontaneous motor features of Parkinsonism, along with a fluctuating cognition, recurrent visual hallucinations of well-formed and detailed imageries. The cognitive decline is severe enough to affect normal social and occupational functioning. In **Parkinson's disease dementia**, affective blunting is prominent often due to well-established Parkinson's disease.

Creutzfeldt-Jakob disease: The clinical picture is one of rapidly deteriorating dementia, myoclonus, cerebellar and extrapyramidal signs leading to death within a year. Patients may present with non-specific symptoms such as lethargy, depression and fatigue. Within weeks, more fulminant symptoms develop, including progressive cortical-pattern dementia, myoclonus and pyramidal and extrapyramidal signs. Myoclonus becomes prominent as the disease progresses. Patients may develop cortical blindness.

The correct answer is: A salient features of Lewy Body dementia → Dementia with prominent but short lucid intervals, A salient feature of CJD → Rapidly progressing dementia, A salient feature of dementia with Parkinson's disease → Emotional blunting

Question 13

Correct

Mark 3.00 out of 3.00

Developmental Psychology

Various psychology and social theories have relevance to psychiatric practice. Which of the following concepts relates most closely to the below descriptions?

1. James Kearney is a 8-year-old boy who has started to demonstrate an awareness of the conservation of objects and inductive reasoning.

Piaget theory of cognitive development ✔

2. Laura Myers is a 4-year-old girl whose mother has been trying to get her to stop snatching toys from others. She has now started to respond to being denied treats if she snatches things from others. Kohlberg's moral development ✔

3. Mrs Neville is a 32-year-old woman who is experiencing periods of despair and protest. Grief ✔

Explanation:

1) The first example describes a child in Piaget's concrete operational stage of cognitive development.

2) The second scenario is an example of moral development, with the child described being in the punishment avoidance stage, as described by Kohlberg.

3) The third vignette describes stages of grief. There are various theories around the stages of grief. Bowlby's 4 stages of grief include protest and despair.

Question 14

Incorrect

Mark 0.00 out of 1.00

Which of the following is a correct statement concerning adjustment disorders?

Select one:

- A stressful life event is not a core feature
- It does not include abnormal grief reactions ✖
- It lasts for about 2 years
- The onset is usually within one month of a stressful life event
- Agitation and dissociation is an important feature

Your answer is incorrect.

Adjustment disorders are conceived of as developing in response to a variety of causal stressful events within a month, the symptoms representing an adaptation to these stressors or their continuing effects. The ICD-10 states that the diagnosis depends on a careful evaluation of the relationship between form, content and severity of symptoms; previous history and personality; and stressful event, situation or life crisis. The latter should be established clearly before the diagnosis can be made, and there should be strong presumptive evidence that the disorder would not have arisen without it.

Answer: The onset is usually within one month of a stressful life event

The correct answer is: The onset is usually within one month of a stressful life event

Question 15

Correct

Mark 1.00 out of 1.00

A 32-year-old man wants to have a surgical operation to look like a woman. The most likely clinical diagnosis is

Select one:

- Dual role transvestism
- Transexualism ✔
- Egodystonic homosexuality
- Body dysmorphophobia
- Fetishistic transvestism

Your answer is correct.

This man has a desire to 'appear like a woman'. This is a telltale feature of transsexualism. In transvestism, cross-dressing is episodic and occurs without a desire to have surgery and other extreme measures to correct sex. Fetishistic transvestites experience sexual arousal.

The correct answer is: Transexualism

Question 16

Correct

Mark 1.00 out of 1.00

Which of the following statements about REM sleep behavioural disorder is incorrect?

Select one:

- It can precede the diagnosis of a movement disorder by several years
- The episodes arise during the middle to latter third of the night during REM sleep
- Treatment with carbamazepine has been shown to be effective ✔
- It is associated with disorders such as Parkinson's disease
- There is no loss of muscle tone and the dreams are acted out as complex behaviours

Your answer is correct.

Normally REM sleep is associated with a loss of muscle tone and dreaming. In RBD, there is no loss of muscle tone, and the dreams are acted as complex behaviours. Patients act out their dreams, with limited awareness of surroundings. The episodes arise during the middle to the latter third of the night during REM sleep. It may occur idiopathically. It is associated with disorders such as Parkinson's disease, diffuse Lewy body disease, multiple system atrophy and Gullian-Barre syndrome. RBD may be the prodrome of neurodegenerative diseases, such as DLB or Parkinson disease. It can precede the diagnosis of a movement disorder by several years. It is likely that the associated lesions are situated in the brainstem. Treatment with clonazepam has been shown to be effective.

The correct answer is: Treatment with carbamazepine has been shown to be effective

Question 17

Correct

Mark 1.00 out of 1.00

Select one atypical feature of schizophrenia

Select one:

- Bizarre delusions
- Mood-congruent delusions
- Systematised delusions
- Mood-incongruent delusions
- Visual but not auditory hallucinations ✔

Your answer is correct.

This is a tricky question. Almost all the listed symptoms have been recognized in schizophrenia. But presenting with visual hallucinations in the absence of auditory hallucinations is rare and must lead to further diagnostic evaluation to rule out organic causes.

The correct answer is: Visual but not auditory hallucinations

Question 18

Correct

Mark 1.00 out of 1.00

An 8-year-old child presents with multiple tics. Which of the following is a diagnostic feature of Tourette's syndrome?

Select one:

- Coprophagia
- Glossolalia
- Vocal tics ✔
- Inattention
- All of the above

Your answer is correct.

Inattention may accompany Tourette's but it is not a diagnostic feature.

The correct answer is: Vocal tics

Question 19

Correct

Mark 1.00 out of 1.00

The original term that described disorganized schizophrenia as a specific diagnostic entity was coined by:

Select one:

- Freud
- Pinel
- Morel
- Hecker ✔
- Sullivan

Your answer is correct.

Hecker coined the term hebephrenia, which predates the DSM concept of disorganized type.

The correct answer is: Hecker

Question 20

Correct

Mark 1.00 out of 1.00

You refer to something the patient said earlier in the interview to move to a new topic. This is called:

Select one:

- Sudden transition
- Smooth transition
- Referred transition ✔
- Distraction technique
- Introduced transition

Your answer is correct.

Referred transition is often used to 'come back' to a topic of importance during a clinical interview.

The correct answer is: Referred transition

Question 21

Correct

Mark 1.00 out of 1.00

A 45-year-old woman has a long-standing history of bipolar disorder and has been on maintenance treatment with lithium carbonate for 8 years. On testing her thyroid functions, her TSH is 6.4 and T4 is normal. What is the next step of management?

Select one:

- Add levothyroxine
- Repeat TSH in a month's time ✔
- Reduce lithium carbonate
- Stop lithium immediately
- Add carbimazole

Your answer is correct.

This lady has a subclinical thyroid disease. This is estimated to be present in 4 to 8.5% of the general population, and up to 20% of women older than 60 years, and about 2% general population are thought to be subclinically hyperthyroid. As she is on lithium, the TSH levels must be monitored in addition to symptoms.

The correct answer is: Repeat TSH in a month's time

Question 22

Correct

Mark 1.00 out of 1.00

An interviewer says, 'I am impressed by what you have achieved despite all these difficulties'. This is best described as

Select one:

- Summation
- Respectful statement ✔
- Inquiry
- Facilitation
- Reflection

Your answer is correct.

Such statements convey the mutual respect maintained by the therapist in the wake of an emotionally important disclosure. This can facilitate further conversations.

The correct answer is: Respectful statement

Question 23

Correct

Mark 1.00 out of 1.00

Visual memory and constructional ability can be tested using:

Select one:

- Sally and Ann test
- Raven's Progressive Matrices
- Rey Auditory Verbal Learning Test
- Rey-Osterrieth Complex Figure Test ✔
- Clock drawing test

Your answer is correct.

Rey-Osterrieth Complex Figure Test (CFT) is a standard measure of perceptual organisation and visual memory in brain-injured individuals.

The correct answer is: Rey-Osterrieth Complex Figure Test

Question 24

Correct

Mark 1.00 out of 1.00

SPECT stands for which of the following?

Select one:

- Single proton emission computerised tomography
- Single positron emission computerised tomography
- Serially produced electronic computerised tomography
- Serially produced electro convulsive therapy
- Single photon emission computerised tomography ✔

Your answer is correct.

Single-photon emission computed tomography (SPECT) is a nuclear tomographic imaging technique that uses gamma rays.

The correct answer is: Single photon emission computerised tomography

Question 25

Correct

Mark 1.00 out of 1.00

Which of the following features is not commonly associated with Parkinson's disease?

Select one:

- Rigidity
- Visual hallucinations
- Bradykinesia
- Spinal flexion ✔
- Postural instability

Your answer is correct.

Hallucinations are seen in nearly 40% of patients with Parkinson's Disease (PD), and fall into three categories:

1. Minor forms, consisting of a sensation of a presence (person), a sideways passage (commonly of an animal) or illusions in 25.5%.
2. Formed visual hallucinations in 22.2%.
3. Auditory hallucinations in 9.7%.

(Ref: Fenelon et al., Brain (2000) 123 (4): 733-745.).

The estimated prevalence of orthostatic hypotension in PD is 30%. (Ref: Velseboer et al. Parkinsonism Relat Disord. 2011;17(10):724-9.)

Camptocormia is the extreme forward flexion of the spine associated with dystonia, Tourette syndrome, amyotrophic lateral sclerosis, and PD. While initially thought to be a rare manifestation of PD, recent estimates of the prevalence of camptocormia in patients with PD vary from 3-12%, making it a relatively uncommon feature of PD. (Ref: Ashour R, Jankovic J. Mov Disord 21:1856-1863, 2006)

The correct answer is: Spinal flexion

Question 26

Correct

Mark 1.00 out of 1.00

A 35-year-old man presents with high blood pressure, rapid pulse and a moist warm skin. His speech is pressured and on examination he has a fine tremor and hyperreflexia. Which of the following drugs has he taken?

Select one:

- Alcohol
- Stimulant ✔
- Opiate
- Hallucinogen
- Inhalant

Your answer is correct.

Stimulants are substances that induce a number of characteristic symptoms. CNS effects include alertness with increased vigilance, a sense of well-being and euphoria. Many users experience insomnia and anorexia, and some may develop psychotic symptoms. Stimulants have peripheral cardiovascular activity, including increased blood pressure and heart rate.

The correct answer is: Stimulant

Question 27

Correct

Mark 1.00 out of 1.00

A 55-year-old businessman presents with headache, poor concentration, memory loss of a fluctuating nature, with unequal pupils and mild papilloedema on examination. The most likely diagnosis is:

Select one:

- Sub-dural haematoma ✔
- Dementia
- Migraine
- Sub-arachnoid haemorrhage
- Motor neurone disease

Your answer is correct.

Subarachnoid haemorrhage generally present more acutely with signs of increased intracranial tension and focal neurological deficits. Chronic SDH (male:female 2:1) is commonly seen in elderly, alcoholics, usually after a mild head injury. The average time between the occurrence of head trauma and the diagnosis of chronic SDH is 4-5 weeks. The presentation is often insidious, with decreased level of consciousness, headache, difficulty with gait or balance, memory loss, mild motor deficits (e.g. hemiparesis) and headache (especially in younger patients).

The correct answer is: Sub-dural haematoma

Question 28

Correct

Mark 1.00 out of 1.00

Which one of the following is an appropriate test for orientation?

Select one:

- Recalling a name and an address
- Knowing today's date ✔
- Subtracting serial 7s from 100
- Knowing the role of the interviewer
- Drawing a clock face

Your answer is correct.

Orientation is usually assessed to time, place and person; it is not particularly sensitive, and intact orientation does not exclude a significant memory disorder, particularly if there is concern about memory from an informant. Time orientation is the most helpful and should include the time of day. Many normal people do not know the exact date, and being out by two days or less is considered normal when scoring this formally.

The correct answer is: Knowing today's date

Question 29

Correct

Mark 1.00 out of 1.00

In a malnourished patient with longstanding alcohol use, which of the following complications could worsen on immediate refeeding with a glucose-rich infusion?

Select one:

- Sodium depletion
- Thiamine deficiency ✔
- Alcoholic hepatitis
- Pancreatitis
- Hypoglycaemia

Your answer is correct.

Thiamine should always be administered prior to glucose infusion because glucose metabolism may rapidly deplete patients' thiamine reserves in cases of long-standing poor nutrition.

The correct answer is: Thiamine deficiency

Question 30

Correct

Mark 1.00 out of 1.00

During psychiatric assessment the best time to carry out an examination of a patient's mental state is

Select one:

- Throughout the interview ✔
- Immediately before discussing diagnosis
- During follow up contact with the patient
- Immediately after presenting complaints
- Immediately after physical examination

Your answer is correct.

Mental state examination takes place throughout the clinical interview.

The correct answer is: Throughout the interview

Question 31

Correct

Mark 1.00 out of 1.00

A 46-year-old woman has depression not responding to antidepressants. She is constipated and has gained weight. Which one is the best screening test for detecting subclinical hypothyroidism?

Select one:

- T4 measurement
- T3 measurement
- Thyroid antibody test
- Thyroid stimulating hormone assay ✔
- Thyroid scan

Your answer is correct.

The best screening test for detecting subclinical hypothyroidism is an assay of TSH hormone levels.

The correct answer is: Thyroid stimulating hormone assay

Question 32

Correct

Mark 1.00 out of 1.00

Which of the following supports a diagnosis of schizophrenia rather than mania?

Select one:

- Hearing voices telling the patient that he is 'powerful'
- Perception of colours as vivid
- Visual hallucinations of small animals
- Hyperacusis
- Hearing voices originating inside the body ✔

Your answer is correct.

Coenaesthetic hallucinations (that emanate from body parts) are suggestive of schizophrenia

The correct answer is: Hearing voices originating inside the body

Question 33

Correct

Mark 1.00 out of 1.00

Choose the most correct statement about the phenomenon of depersonalization seen in several psychiatric disorders.

Select one:

- It is easily distinguished from derealisation
- Insight is often impaired
- It is an objective experience
- It is associated with an unpleasant emotional state ✔
- It is not experienced by people without any mental health issues

Your answer is correct.

Depersonalization is an unpleasant subjective experience where the patient feels as if they have become 'unreal'. It is a non-specific symptom occurring in many psychiatric disorders as well as in healthy people during periods of fatigue, stress, etc.

The correct answer is: It is associated with an unpleasant emotional state

Question 34

Correct

Mark 1.00 out of 1.00

"Mrs. X affirms she has no brain, no nerves, no chest, no stomach, no intestines; there's only skin and bones of a decomposing body. She has no soul, and she is nothing more than a decomposing body". This description is characteristic of:

Select one:

- Somatoform disorder
- Nihilistic delusions ✔
- Hypochondriasis
- Factitious disorder
- Alexithymia

Your answer is correct.

Patients with psychotic depression may exhibit nihilistic delusions where they deny the normal functioning and existence of their body, self, others and the world around them. The example in this question is a translated excerpt from Cotard's original case of a 43-year-old woman described in 1880.

(Ref: Debruyne et al. Mind & Brain:2011; 2(1))

The correct answer is: Nihilistic delusions

Question 35

Correct

Mark 1.00 out of 1.00

Which of the following describes a delusional perception?

Select one:

- Stimulus in one modality experienced in another
- No stimulus, abnormal perception
- Normal perception, delusional meaning ✔
- No stimulus, delusional meaning
- Abnormal perception, abnormal meaning

Your answer is correct.

Delusional perception is a primary delusion, which arises as a result of a normal perception. The percept is a real external object and not a hallucinatory experience.

The correct answer is: Normal perception, delusional meaning

Question 36

Correct

Mark 1.00 out of 1.00

Obsessional slowness is common in which of the following types of obsessions?

Select one:

- Washing obsessions
- Obsession of symmetry ✔
- Cleaning obsessions
- Religious obsessions
- Obsessions of contamination

Your answer is correct.

An obsessive need for symmetry and exactness and a compulsion to sort and arrange results in slowness.

The correct answer is: Obsession of symmetry

Question 37

Correct

Mark 1.00 out of 1.00

Which of the following pertains to human research ethics?

Select one:

- Birmingham code
- NICE Guidelines
- Newcastle declaration
- Nuremberg code ✔
- None of the above

Your answer is correct.

The interest in research ethics was stimulated by the atrocities committed during Second World War. Numerous unethical experiments including forceful sterilization, treacherous poisoning, inducing malaria and typhus, high-altitude experiments and severe dehydration were conducted by Nazi doctors on prisoners at concentration camps. A series of trials took place at Nuremberg following the world war and a code for human experimentation was set up. This is popularly called as Nuremburg code.

The correct answer is: Nuremberg code

Question 38

Correct

Mark 1.00 out of 1.00

Which of the following culture-bound syndromes literally means 'nerve weakness'?

Select one:

- Shenjing shuairuo ✔
- Dhat
- Mal de mojo
- Koro
- Bouffée délirante

Your answer is correct.

Shenjing shuairuo is a culture-bound syndrome seen in China. Literally means "nerve weakness." Symptoms include - chronic fatigue, dizziness, headaches, sleep difficulties, nonspecific, and multiple somatic complaints. These conditions are likely related to depression, anxiety, fibromyalgia, chronic fatigue, and irritable bowel syndromes.

The correct answer is: Shenjing shuairuo

Question 39

Correct

Mark 1.00 out of 1.00

Social role valorisation is best described as

Select one:

- Loss of social role with advancing age
- An illness behaviour
- The basis of human interaction within a cost benefit framework
- A concept opposite to normalisation
- Social acceptance of people with disabilities without devaluation ✔

Your answer is correct.

Social role valorisation is best described as a concept formulated in 1983 by Wolf Wolfensberger as the successor to the principle of normalization that deals with the enablement, establishment, enhancement, maintenance, and/or defense of valued social roles for people. SRV is primarily a response to the historically universal phenomenon of social devaluation and especially societal devaluation.

The correct answer is: Social acceptance of people with disabilities without devaluation

Question 40

Incorrect

Mark 0.00 out of 1.00

In ICD-11, which of the following groupings includes a dimensional element for the diagnoses within it?

Select one:

- Dissociative disorders
- Disorders due to substance use and addictive disorders
- Schizophrenia and other primary psychotic disorders
- Anxiety and fear related disorders ✖
- Catatonia

Your answer is incorrect.

'Personality disorders', 'schizophrenia and other primary psychotic disorders', and depressive episodes in 'mood disorders' each include significant dimensional elements to enable clearer descriptions of the symptomatology experienced by patients. In schizophrenia and other primary psychotic disorders, this includes: positive symptoms; negative symptoms; depressive mood symptoms; manic mood symptoms; psychomotor symptoms; and cognitive symptoms. Neurodevelopmental disorders also include qualifiers for a diagnosis of autism spectrum disorder to enable a description of the extent of impairment of intellectual functioning and language abilities.

The correct answer is: Schizophrenia and other primary psychotic disorders

Question 41

Correct

Mark 1.00 out of 1.00

A 30-year-old couple move from India to the UK. The man is a doctor and his wife is a teacher. They have two children who are pre-school. The couple soon settles down in a friendly neighbourhood. Later, the husband's parents join the family. Who is most likely to develop biculturalism?

Select one:

- The neighbours
- The husband
- The grandparents
- The wife
- The children ✔

Your answer is correct.

Children adapt more easily and develop biculturalism more frequently than adults and the elderly when immigrating.

The correct answer is: The children

Question 42

Correct

Mark 1.00 out of 1.00

Cultural formulation includes all of the following except:

Select one:

- Acknowledging cultural elements in the patient-therapist relationship
- The therapist's view of the patient's cultural identity ✔
- Considering cultural explanations of exhibited distress
- Identifying the influence of the psychosocial environment on disease presentation
- Making diagnosis in the context of patient's background culture

Your answer is correct.

The patient's own view of his cultural identity is more important than the therapist's view.

The correct answer is: The therapist's view of the patient's cultural identity

Question 43

Correct

Mark 1.00 out of 1.00

Who raised the question "Is there such a thing as mental illness?" and argued against the existence of mental illnesses?

Select one:

- Thomas Hardy
- Jacques Lacan
- Goffmann
- Foucault
- Thomas Szasz ✔

Your answer is correct.

Thomas Szasz, a prominent anti-psychiatrist, argued against the existence of mental illness. The book 'Myth of mental illness' was written by him.

The correct answer is: Thomas Szasz

Question 44

Correct

Mark 1.00 out of 1.00

Which of the following determines one's cultural identity?

Select one:

- Upbringing and choice ✔
- Place of birth
- Attitudes and behaviour
- Language and knowledge
- Socioeconomic restraints

Your answer is correct.

Race is perceived as a permanent entity determined by genetic ancestry and characterised by physical appearance.

Culture is perceived as a changeable entity determined by upbringing and choice and characterised by behaviours and attitudes.

Ethnicity is perceived as partially changeable entity determined by social pressure and psychological need characterised by a sense of group identity and belonging.

(Ref: Seminars in General Adult Psychiatry, RCPsych Gaskell Press, Pg 783.)

The correct answer is: Upbringing and choice

Question 45

Correct

Mark 1.00 out of 1.00

The right action is the one that has the best foreseeable consequences. This is known as:

Select one:

- Engel's model
- Paternalistic model
- Utilitarian moral theory ✔
- Hammurabi code
- Oath of Hippocrates

Your answer is correct.

Consequentialist moral theories are teleological: they aim at some goal state and evaluate the morality of actions in terms of progress toward that state.

The best-known version of consequentialism is **utilitarianism**. As most clearly stated by Mill (who was a proponent of the utilitarianism), the basic principle of utilitarianism is: *Actions are right to the degree that they tend to promote the greatest good for the greatest number.*

In contrast, deontological principles are duty-based. Kant's theory is an example of a deontological or duty-based ethics: it judges morality by examining the nature of actions and the will of agents rather than goals achieved. A deontological theory looks at inputs rather than outcomes.

The correct answer is: Utilitarian moral theory

Question 46

Correct

Mark 1.00 out of 1.00

Which of the following studies are associated with deliberate infection in people with learning disabilities?

Select one:

- Belmont Study
- Willowbrook School Study ✔
- Nuremberg Study
- Tuskegee Study
- Jewish Chronic Disease Hospital Study

Your answer is correct.

Willowbrook School Study (1963 - 1966): Willowbrook State School, situated in New York State, was an institution for mentally handicapped children. Parents of children in the institution gave consent for their children to participate in a study. The intent of the research study was to follow the course of viral hepatitis and study the effectiveness of an agent for inoculating against hepatitis. Parents were provided with study information describing the drug administration as vaccinations. However, the children were deliberately infected with the hepatitis virus. There is evidence that the school only admitted children to the school whose parents gave permission for them to be in the study.

Jewish Chronic Disease Hospital (1963): Studies were conducted at the Jewish Chronic Disease Hospital in New York City to develop information on the nature of the human transplant rejection process. Chronically ill patients who did not have cancer were injected with live human cancer cells. The physicians did not inform the patients as to what they were doing. The physicians' rationalization for their actions was as follows: (i) they did not want to scare the patients and (ii) they thought the cells would be rejected. (http://iris.uwaterloo.ca/ethics/human/resources/index.htm)

The correct answer is: Willowbrook School Study

Question 47

Correct

Mark 1.00 out of 1.00

Which of the following is a new mental health disorder under ICD-11?

Select one:

- Trichitillomania
- Prolonged grief disorder ✔
- Sadomasochistic disorder
- Hyperkinetic disorder
- Acute stress reaction

Your answer is correct.

Prolonged grief disorder is a new mental health disorder under ICD-11 categorised under disorders specifically associated with stress. It describes a persistent and disabling grief reaction, which lasts longer than 6 months and is out of keeping with what is expected for the cultural context.

'Trichotillomania' was in ICD-10, as was hyperkinetic disorder, which has now been re-categorised as ADHD.

Sadomasochism and other consensual sexual practices are no longer generally considered to be pathological.

'Acute stress reaction' is no longer seen as a mental health condition, but as a normal reaction to extreme stress. It is categorised under factors influencing health status or contact with health services.

The correct answer is: Prolonged grief disorder

Question 48

Correct

Mark 1.00 out of 1.00

Fear of turning into a cannibal is one of the important features of which culture-bound syndrome?

Select one:

- Latah
- Amok
- Dhat syndrome
- Windigo ✔
- Koro

Your answer is correct.

In Windigo, a fear of turning into a demon or cannibal is often seen.

The correct answer is: Windigo

Question 49

Correct

Mark 1.00 out of 1.00

Tom has history of severe alcohol dependence. He has been given disulfiram to reduce the frequency of drinking alcohol. This is an example of:

Select one:

- Aversive conditioning ✔
- Premack's principle
- Covert reinforcement
- Avoidance learning
- Covert sensitization

Your answer is correct.

Aversive conditioning: This is an operant conditioning where punishment is used to reduce the frequency of target behaviour e.g. the use of Disulfiram (noxious stimuli) to reduce the frequency of drinking alcohol.

The correct answer is: Aversive conditioning

Question 50

Correct

Mark 1.00 out of 1.00

Timmy has a pet mouse which he spends time with. As part of a psychological experiment, his father creates a loud noise whenever Timmy is playing with his mouse, and the noise frightens Timmy. This is done a number of times, so that whenever Timmy sees his pet mouse, he feels frightened. Which of these is now classed as the conditioned stimulus?

Select one:

- The mouse ✔
- The noise
- The mouse + noise
- The fear
- The fear + noise

Your answer is correct.

The mouse was previously a neutral stimulus; it has been associated with an unconditioned stimulus (the noise) that naturally triggers an automatic response in Timmy (fear). The mouse therefore becomes a conditioned stimulus, eliciting fear in Timmy.

The correct answer is: The mouse

Question 51

Correct

Mark 1.00 out of 1.00

A psychologist is involved in treating a patient with fear of heights. He requested the patient to imagine standing on top of the building without actual real exposure to them. The psychological principle employed here is:

Select one:

- Implosion ✔
- Reciprocal inhibition
- Punishment
- Aversion
- Flooding

Your answer is correct.

Flooding is an operant conditioning technique where actual exposure to feared stimulus takes place for a substantial amount of time so the accompanying anxiety response fades away while the stimulus is continuously present. This will lead to extinction of fear. When a similar technique is attempted with imagined exposure then it is called **implosion**.

The correct answer is: Implosion

Question 52

Correct

Mark 1.00 out of 1.00

There is no processing involved in which of the following types of memory?

Select one:

- Recent memory
- Short term memory
- Sensory memory ✔
- Remote memory
- Long term memory

Your answer is correct.

Sensory memory: This is modality specific, has a large capacity but gets disrupted by an inflow of new information in the same modality.

Each sense has its own sensory memory e.g. iconic (visual), echoic (auditory). Iconic lasts for 0.5 seconds while echoic lasts for 2 seconds.

No processing is involved in sensory memory. Information that is attended to is transferred to short-term memory store, and information that is not selected for processing is rapidly lost.

The correct answer is: Sensory memory

Question 53

Correct

Mark 1.00 out of 1.00

All are examples of secondary drive except

Select one:

- Dominance
- Comfort
- Achievement
- Sex ✔
- Socialisation

Your answer is correct.

Drives refer to purposeful activity initially produced to satisfy an internal need. Drives are commonly divided into primary and secondary drives.

Examples of primary drives are eating, drinking, sex.

Examples of secondary drives are affection, comfort, achievement, dominance and socialisation. The secondary drives acquire their values through their association with primary drives and often through a social learning process.

The correct answer is: Sex

Question 54

Correct

Mark 1.00 out of 1.00

A child received generous praise from her parents every time when she wrote a letter of the alphabet. She was praised more when she managed to write a full word. She was rewarded more generously when she wrote a sentence. Which of the following reinforcement is taking place?

Select one:

- Incubation
- Negative reinforcement
- Covert reinforcement
- Shaping ✔
- Sensitisation

Your answer is correct.

Shaping is an operant conditioning method for producing an entirely new behaviour by using rewards to guide an organism toward a desired behaviour (called 'successive approximations'). The organism is rewarded with each small advancement in the right direction.

The correct answer is: Shaping

Question 55

Correct

Mark 1.00 out of 1.00

Which of the following is one of the maladaptive traits that can be used to further describe a diagnosis of personality disorder in ICD-11?

Select one:

- Mood variability
- Impulsivity
- Dissociality ✔
- Affective lability
- Anxiousness

Your answer is correct.

The maladaptive traits that can be used to describe a diagnosis of personality disorder in ICD-11 are: negative affectivity, detachment, dissociality, disinhibition and anankastia. A further qualifier of 'borderline pattern' can also be used.

The correct answer is: Dissociality

Question 56

Correct

Mark 1.00 out of 1.00

Lucy is learning to write. Her father praises her for holding the pencil correctly, then for scribbling something on a piece of paper and writing something vaguely which resembles like a letter. The learning theory principle employed here is:

Select one:

- Chaining
- Positive reinforcement
- Shaping ✔
- Stimulus generalization
- Stimulus discrimination

Your answer is correct.

Shaping is a teaching technique by which a child is rewarded for a successful approximation of a target skill. If a child is learning to write his or her name, say "Alex," a teacher would shape the responses by starting with the letter A and providing the other letters: _lex. The next step would be perhaps for the child to write the first and last letters independently. The process would continue until the child can write his name independently.

(Webster J, 2008. Retrieved from http://specialed.about.com/od/glossary/g/shaping.htm)

The correct answer is: Shaping

Question 57

Correct

Mark 1.00 out of 1.00

Which of the following concepts is reflective of a change in ICD-11 (from ICD-10)?

Select one:

- Use of dimensional approach for all disorders
- Removal of minimum symptom count for all disorders
- Introduction of gender incongruence as mental health disorder
- Cultural considerations have been removed
- All disorders can be diagnosed at any age ✔

Your answer is correct.

ICD-11 takes a lifespan approach to diagnosis, permitting diagnoses which were usually associated with childhood or adolescence, such as attachment or conduct disorder, to be diagnosed at any age. It specifically considers the cultural background of a patient in diagnosis. A dimensional approach has been taken in personality and primary psychotic disorders, as well as for depressive episodes in mood disorders. Minimum symptom counts have been removed from diagnostic criteria for most disorders, but not major depression. Gender incongruence, replacing gender identity disorder, has been moved to the sexual health chapter of ICD-11 as it is not a mental health disorder.

The correct answer is: All disorders can be diagnosed at any age

Question 58

Correct

Mark 1.00 out of 1.00

Who proposed the two-factor theory of intelligence?

Select one:

- Lazarus
- James-Lange
- Sternberg
- Spearman ✔
- Maslow

Your answer is correct.

The two-factor theory of intelligence was postulated by **Spearman**. Spearman carried out a factor analysis of the result of children's performance on a number of tests and concluded that all tests measured both a common factor of general intelligence (g) and a specific factor (s). He believed that individual differences were due to differences in (g).

According to **Sternberg**, component intelligence is used for executive tasks and experiential intelligence is used for routine tasks that have already been learnt or mastered.

The correct answer is: Spearman

Question 59

Correct

Mark 1.00 out of 1.00

A 22-year-old man is most likely to fall under which stage of Erikson's psychosocial development?

Select one:

- Initiative vs. guilt
- Identity vs. role confusion
- Industry vs. inferiority
- Integrity vs. despair
- Intimacy vs. isolation ✔

Your answer is correct.

During young adulthood (between the ages of 20 and 40), a life crisis of intimacy vs. isolation occurs. The outcomes are love and commitment/superficial relationships.

The correct answer is: Intimacy vs. isolation

Question 60

Correct

Mark 1.00 out of 1.00

In Piaget's theory the development of object permanence occurs during which of the following stages?

Select one:

- Sensorimotor stage ✔
- Preoperational stage
- None of the above
- Formal operational stage
- Concrete operational stage

Your answer is correct.

Jean Piaget, a Swiss psychologist, believed that intelligence is the product of a natural and inevitable sequence of developmental stages. He did not suggest that it was simply born, but rather that it developed as a result of the constantly changing interaction of the child and the environment.

He also believed that the child went through different stages of development, and each stage is characterised by qualitatively different ways of thinking about the world. The earliest of these stages is the sensorimotor stage (birth to 2) when the baby differentiates self from objects, recognises self as agent of action and begins to act intentionally. The baby also develops object permanence i.e. an object that is not seen can continue to exist.

The correct answer is: Sensorimotor stage

Question 61

Correct

Mark 1.00 out of 1.00

A child who is observed to have a 'secure attachment pattern' in the strange situation experiment will show which of the following?

Select one:

- Shows discomfort for being alone and not for mother's absence
- Rejecting, hostile and intrusive
- Terrified and frozen when the mother returns
- Shows discomfort when mother leaves but not for being alone ✔
- Strongly resists mother's attempts at reconciliation

Your answer is correct.

In secure attachment pattern, the child uses the mother as a secure base, explores freely, may be distressed at separation but greets positively on the reunion, seeks comfort, settles down and returns to exploration. Secure infants anticipate that their caregiver is accessible and available to them for comfort when necessary, and their primary carer is sensitive to the infants' signals and cues.

(Ref: Thambirajah, MS. Psychological basis of psychiatry. Pg: 282)

The correct answer is: Shows discomfort when mother leaves but not for being alone

Question 62

Correct

Mark 1.00 out of 1.00

Which of the following is a risk factor for abuse during childhood?

Select one:

- Single parent family ✔
- Mother's age more than 28
- Internal conflicts that are resolved in the family
- Family having a network of frequently visiting friends
- Family with children aged older than 7

Your answer is correct.

The rate of child abuse in single parent households is 27.3 children per 1,000, which is nearly twice the rate of child abuse in two-parent households (15.5 children per 1,000).

(Goldman et al., 2003)

The correct answer is: Single parent family

Question 63

Correct

Mark 1.00 out of 1.00

All of the following are risk factors that could be targeted to reduce delayed language development in children except

Select one:

- Smoking in pregnancy
- Consumption of alcohol in pregnancy
- Behavioural problems in the child
- Inadequate cognitive stimulation
- Small family size ✔

Your answer is correct.

A large sized, overcrowded family is associated with more delays in language development. (CR104: RCPsych)

The correct answer is: Small family size

Question 64

Correct

Mark 1.00 out of 1.00

Which of the following theories explains how boys 'learn to become boys' and girls 'learn to become girls'?

Select one:

- Insight learning
- Rule learning
- Concept learning
- Latent learning
- Social learning ✔

Your answer is correct.

Social learning theorists identified specific sex-typed behaviours (Mischel, 1970). A behaviour is sex-typed when it is more expected and therefore seen as appropriate when performed by one sex, but less expected and therefore seen as inappropriate when performed by the other sex. Gender socialisation works, according to social learning theorists, by rewarding children for engaging in sex-typed behaviour that is consistent with their assigned sex category. The classic example is crying; while a little girl may be soothed when she cries, a little boy may be told that boys don't cry. Crying is a sex-typed behaviour, seen as OK for girls and therefore not a punishable behaviour. But because it is not seen as an appropriate behaviour for boys, the little boy may be punished or corrected for his crying behaviour. Through these kinds of interactions, gender socialisation occurs.

(www.sagepub.com/upm-data/39367_4.pdf)

The correct answer is: Social learning

Question 65

Incorrect

Mark 0.00 out of 1.00

The social behaviour of buying dolls as gifts for girls and cars for boys is linked to which of the following concepts?

Select one:

- Imprinting
- Social learning
- Self-realisation
- Gender schema ✖
- Cognitive mapping

Your answer is incorrect.

According to the social learning theory, a child learns several gender-related behaviours through reinforcement and modeling (Golombok & Fivush, 1994). This is acquired first through parents who give the child his or her initial social position. Parents begin this process by typically encouraging sons to be more independent, competitive and achieving than they do for daughters (Lamb, 1986; Block, 1979). Parents also encourage daughters to be more passive and to seek protection (Chafetz, 1978). There have been differences shown in the reinforcement fathers and mothers give to sons and daughters. Even parents who strive to be egalitarian in their dealings with their children have been found to discourage certain non-traditional play behaviours more in one gender than the other. For example, parents are more likely to discourage their sons from playing with dolls than they are to discourage their daughters from playing with trucks (Weisner & Wilson-Mitchell, 1990). This behaviour on the part of parents (and others in the society) serves to model and reinforce gender stereotyped behaviours in children.

(For more info on socialization theory refer to http://www3.uakron.edu/witt/father/parke5.htm)

The correct answer is: Social learning

Question 66

Correct

Mark 1.00 out of 1.00

'A child studies inner and outer world via senses but perceives mother and self as one unit'. This refers to which stage of Mahler's theory?

Select one:

- Symbiosis ✔
- Practising sub-phase of separation-individuation
- Differentiation sub-phase of separation-individuation
- Normal autism
- Rapprochement sub-phase of separation-individuation

Your answer is correct.

The stage of normal autism lasts from 0 to 2 months. Here the child spends most time in sleep, as if the intrauterine aloofness continues.

The stage of **symbiosis** lasts from 2 to 5 months. Here, inner and outer world is studied by the child via senses but perceives mother and self as one unit.

The correct answer is: Symbiosis

Question 67

Incorrect

Mark 0.00 out of 1.00

What is the incidence of fatal hepatotoxicity with valproate?

Select one:

- 1 in 250
- 1 in 100
- 1 in 100,000
- 1 in 1,000
- 1 in 10,000 ✖

Your answer is incorrect.

In those taking valproate alone: 1 in 100,000 patients.

Risk factors for hepatotoxicity include young age (less than ten years old), sensitivity during the first few months of treatment, severe epilepsy, and use of multiple drugs.

The correct answer is: 1 in 100,000

Question 68

Correct

Mark 1.00 out of 1.00

A 65-year-old man develops recurrent UTI since starting an antidepressant. He also gives a history suggestive of new onset prostatism. But prostate examination shows no enlargement. The most likely offending agent is:

Select one:

- Nefazadone
- Reboxetine
- Trazodone
- Mirtazapine
- Amitryptyline ✔

Your answer is correct.

Drugs with significant anticholinergic properties such as amitriptyline notoriously cause signs of prostatism in elderly males. They may produce urinary retention and sometimes bladder distension.

The correct answer is: Amitryptyline

Question 69

Correct

Mark 1.00 out of 1.00

Tom is an inpatient on a psychiatric ward. He has a history of psychotic depression and is on combination treatment with antipsychotics and antidepressants. He appears slightly flushed, with dry mouth, dry skin, feeling hot, and confused. He is exhibiting features of:

Select one:

- SSRI discontinuation syndrome
- Neuroleptic malignant syndrome
- Idiopathic parkinsonism
- Anticholinergic syndrome ✔
- Serotonin syndrome

Your answer is correct.

Anticholinergic syndrome results from the inhibition of cholinergic neurotransmission at muscarinic receptor sites. Psychotropics such as phenothiazines, tricyclic antidepressants and antiparkinsonian agents have anticholinergic properties. When they are administered in combination, the action of these drugs becomes additive resulting in anticholinergic syndrome. Features of a mild organic brain syndrome, which includes disorientation, difficulty in concentrating and impaired short-term memory are seen, especially noticeable at nighttime. Other physical symptoms include flushing, dry skin and mucosa, mydriasis with loss of accommodation and fever.

(Ref: Kaplan & Sadock Synopsis of Psychiatry 9th ed. Pg 1170-1172)

The correct answer is: Anticholinergic syndrome

Question 70

Correct

Mark 1.00 out of 1.00

Patients on tranylcypromine should not take an excess of

Select one:

- Swede
- Banana
- Cheese ✔
- Potatoes
- Salt

Your answer is correct.

Cheese can induce tyramine related reaction in those on MAOIs.

The correct answer is: Cheese

Question 71

Correct

Mark 1.00 out of 1.00

Which of the following antipsychotics is associated with photosensitivity more than others listed?

Select one:

- Olanzapine
- Quetiapine
- Haloperidol
- Chlorpromazine ✔
- Risperidone

Your answer is correct.

Allergic dermatitis and photosensitivity can occur with low-potency agents. Long-term chlorpromazine use can cause blue-gray discoloration of skin areas exposed to sunlight. This is reversible.

Irreversible retinal pigmentation is associated with the use of high dose thioridazine (above 1000 mg a day).

The correct answer is: Chlorpromazine

Question 72

Incorrect

Mark 0.00 out of 1.00

A 56-year-old gentleman suffers from a severe depressive episode. He has a history of bipolar affective disorder. He is on maintenance treatment with Lithium for five years. What is the most appropriate drug to treat his depression?

Select one:

- Amitriptyline
- Venlafaxine ✖
- Phenelzine
- Citalopram
- Duloxetine

Your answer is incorrect.

The risk of inducing a switch to mania is less with SSRIs and more with other classes of antidepressants such as TCAs, MAOIs and SNRIs.

The correct answer is: Citalopram

Question 73

Incorrect

Mark 0.00 out of 1.00

Which of the following drugs is used in the treatment of akathisia?

Select one:

- Procyclidine
- Atrophine
- Trihexyphenidyl ✖
- Clonazepam
- Benperidol

Your answer is incorrect.

Treatments for akathisia include reducing the antipsychotic drug or change to an atypical drug. A reduction in symptoms of akathisia may be seen with propranolol, clonazepam, mirtazapine, trazodone, mianserin and cyproheptadine.

The correct answer is: Clonazepam

Question 74

Correct

Mark 1.00 out of 1.00

Which of the following drugs can cause severe rebound hypertension when stopped?

Select one:

- Risperidone
- Venlafaxine
- Duloxetine
- Paroxetine
- Clonidine ✔

Your answer is correct.

Clonidine is an alpha-2 adrenergic receptor agonist. Agonist effects at these receptors, which are located presynaptically in adrenergic neurons, reduce the release of epinephrine from these neurons.

The correct answer is: Clonidine

Question 75

Correct

Mark 1.00 out of 1.00

Tom is a 10-year-old boy with Tourettes' syndrome. He has had a medication for treating his condition, which has lead to significant weight gain. Choose the best alternative?

Select one:

- Risperidone
- Clonidine ✔
- Haloperidol
- Pimozide
- Olanzapine

Your answer is correct.

Weight gain may be more pronounced in children than adults with antipsychotics and this may limit their use. Clonidine has been shown in open studies to reduce the severity and frequency of tics and could bring a substantial reduction in tics.

(Ref; Leckman JF et al.; Arch Gen Psychiatry;1991: 42: 324-328)

The correct answer is: Clonidine

Question 76

Correct

Mark 1.00 out of 1.00

Which of the following statement regarding lithium is NOT true?

Select one:

- Oral preparations may have different bioavailability
- Hypothyroidism is more common in middle-aged women taking lithium
- ECG changes can be seen even at therapeutic doses
- Coarse tremor is a common side effect ✔
- Peripheral limb oedema is a recognised side effect of lithium use

Your answer is correct.

Lithium chloride causes a concentration dependent block of the peak sodium current in cardiac pacemakers, but even at therapeutic doses, abnormal ECG can be documented, often with no clinical implications.

Fine hand tremor is a common side effect often seen at lower doses; coarse tremor indicates toxicity.

The correct answer is: Coarse tremor is a common side effect

Question 77

Correct

Mark 1.00 out of 1.00

Co-agonists glycine and D-serine act on which of the following receptors?

Select one:

- AMPA ✔
- Serotonergic
- Adenosine
- Muscarinic
- Kainate

Your answer is correct.

D-serine is an endogenous ligand for the glycine site of the N-methyl-D-aspartate receptor. Some recent studies suggest that there is a low affinity and stereo-selective site at the agonist-binding pocket of AMPA receptors as well for D-amino acids such as d-serine.

The correct answer is: AMPA

Question 78

Correct

Mark 1.00 out of 1.00

The constituents of khat have action that has greatest similarity to which of the following drugs?

Select one:

- Cocaine
- Phencyclidine
- Amphetamine ✔
- LSD
- Cannabis

Your answer is correct.

The fresh leaves of Catha edulis from East Africa were used as a stimulant for amphetamine-like effects from the main ingredient cathinone. Cathinone has most of the CNS and peripheral actions of amphetamine. It is typically absorbed buccally after chewing the leaf and being an alkaloid, gets metabolized rapidly with low chances of toxicity.

The correct answer is: Amphetamine

Question 79

Correct

Mark 1.00 out of 1.00

A 56-year-old woman on treatment with clozapine has some residual symptoms and a BMI of 27. The ideal drug of choice for adjunctive treatment is:

Select one:

- Olanzapine
- Citalopram
- Fluoxetine
- Quetiapine
- Aripiprazole ✔

Your answer is correct.

The addition of **aripiprazole** to clozapine may be particularly effective in reversing metabolic effects (such as weight gain) and could be used as an augmenting agent to clozapine.

The correct answer is: Aripiprazole

Question 80

Correct

Mark 1.00 out of 1.00

The mechanism of action of cyproterone acetate is:

Select one:

- Blocks testosterone receptors ✔
- LHRH antagonism
- Reduces prolactin levels
- Agonist at testosterone receptors
- 17- alpha-hydroxycatabolism

Your answer is correct.

The mechanism of action of cyproterone acetate is to **block testosterone receptors**. The direct antiandrogenic effect of cyproterone is to block the binding of dihydrotestosterone to the specific receptors in the prostatic carcinoma cell. In addition, cyproterone exerts a negative feed-back on the hypothalamo-pituitary axis, by inhibiting the secretion of LH resulting in diminished production of testicular testosterone.

(Retrieved from Archives of Product Info: http://www.drugs.com/mmx/cyproterone-acetate.html).

The correct answer is: Blocks testosterone receptors

Question 81

Correct

Mark 1.00 out of 1.00

Which of the following drugs acts as a partial mu receptor agonist?

Select one:

- Acamprosate
- Buprenorphine ✔
- Disulfiram
- Methadone
- Buspirone

Your answer is correct.

Buprenorphine is a partial agonist at the mu-opioid receptor and has a reduced risk of fatal respiratory depression.

The correct answer is: Buprenorphine

Question 82

Correct

Mark 1.00 out of 1.00

Which of the following atypical antidepressants have dopaminergic and noradrenergic actions?

Select one:

- Bupropion ✔
- Mirtazapine
- Venlafaxine
- Reboxetine
- Mianserin

Your answer is correct.

Bupropion is an atypical antidepressants with dopaminergic and noradrenergic actions, and it has been advocated by NICE for smoking cessation.

The correct answer is: Bupropion

Question 83

Correct

Mark 1.00 out of 1.00

The drug used in treatment of REM sleep behavioural disorder is

Select one:

- Citalopram
- Sildenafil
- Amitryptyline
- Olanzapine
- Clonazepam ✔

Your answer is correct.

Treatment with **clonazepam** has shown to be effective in REM sleep behavioural disorders.

The correct answer is: Clonazepam

Question 84

Correct

Mark 1.00 out of 1.00

Which of the following acts on presynaptic receptors as main mechanism of action?

Select one:

- Phenelzine
- Propronalol
- Donepezil
- Diazepam
- Clonidine ✔

Your answer is correct.

Clonidine is a sympatholytic medication, which is classified as a centrally acting a2-adrenergic agonist. It has specificity towards the presynaptic a2 receptors in the vasomotor center in the brainstem.

The correct answer is: Clonidine

Question 85

Correct

Mark 1.00 out of 1.00

A patient has chronic schizophrenia and is on clozapine therapy. If you want to try augmentation, which one should be avoided?

Select one:

- Carbamazepine ✔
- Sodium valproate
- Aripiprazole
- Fluoxetine
- Lithium

Your answer is correct.

The combination of **carbamazepine and clozapine** is generally **contraindicated**. First, carbamazepine and clozapine each produce hematologic side effects at higher frequencies than do most other available drugs. Another consideration is that carbamazepine lowers the level of clozapine, possibly more than it lowers levels of traditional neuroleptics.

Am J Psychiatry 155:12-21, January 1998

The correct answer is: Carbamazepine

Question 86

Incorrect

Mark 0.00 out of 1.00

Liz was started on a mood stabiliser when she was diagnosed with bipolar affective disorder type 2. Her psychiatrist says that the dose needs to be carefully titrated as the prescribed mood stabiliser induces its own metabolism. Which of the following has been prescribed to Liz?

Select one:

- Carbamazepine
- Semisodium valproate ✖
- Lamotrigine
- Lithium
- Sodium valproate

Your answer is incorrect.

Carbamazepine has a tendency to induce its own metabolism leading to reduced plasma steady state levels.

The correct answer is: Carbamazepine

Question 87

Correct

Mark 1.00 out of 1.00

Which of the following does not increase tricyclic antidepressant concentrations?

Select one:

- Cigarette smoking ✔
- Haloperidol
- Quetiapine
- Methylphenidate
- Risperidone

Your answer is correct.

Antipsychotics, methylphenidate, acetazolamide, aspirin, cimetidine, thiazides, fluoxetine and sodium bicarbonate increase tricyclic antidepressant (TCA) concentrations. Cigarette smoking decreases their concentration through its action on the CYP1A2 enzyme.

The correct answer is: Cigarette smoking

Question 88

Incorrect

Mark 0.00 out of 1.00

The antidepressant with the longest mean half-life from the given list is:

Select one:

- Fluvoxamine
- Venlafaxine
- Citalopram
- Paroxetine ✖
- Trazodone

Your answer is incorrect.

Fluoxetine has the longest half-life of all the SSRIs, one to three days, because of a metabolite.

Paroxetine, the SSRI with a long t1/2 compared to the others, has a half-life of only 21 hours.

The correct answer is: Citalopram

Question 89

Correct

Mark 1.00 out of 1.00

A patient who is prescribed disulfiram to aid alcohol abstinence, presents to Accident and Emergency room with facial flushing, hypotension, tachycardia, nausea, and vomiting. She denies any recent ingestion of alcohol. Which of the following could have caused her symptoms?

Select one:

- Walnuts
- Mature cheese
- Banana
- Chocolate
- Cough syrup ✔

Your answer is correct.

Cough and cold medications bought over the counter often contain alcohol and can precipitate a disulfiram reaction. Severe Antabuse reaction (also known as disulfiram reaction) is characterised by respiratory depression, seizures, cardiovascular collapse, and even death.

The correct answer is: Cough syrup

Question 90

Correct

Mark 1.00 out of 1.00

A barbiturate-dependent patient shows little pharmacological response to administration of benzodiazepines. This is called:

Select one:

- Tolerance
- Dependence
- Downregulation
- Sensitisation
- Cross-tolerance ✔

Your answer is correct.

Cross-tolerance refers to decreased response to one medication due to exposure to a different medication. Here exposure to barbiturates has produced cross-tolerance to benzodiazepines.

The correct answer is: Cross-tolerance

Question 91

Correct

Mark 1.00 out of 1.00

The benzodiazepine with the longest half life in a young adult is:

Select one:

- Alprazolam
- Diazepam ✔
- Clonazepam
- Nitrazepam
- Temazepam

Your answer is correct.

Diazepam is a benzodiazepine which has a very long elimination half-life (20 - 100 hours with an active metabolite 36 - 200 hours); as a result withdrawal symptoms are not very common with diazepam.

The correct answer is: Diazepam

Question 92

Incorrect

Mark 0.00 out of 1.00

When administered orally, the benzodiazepine that produces the most rapid onset of action is:

Select one:

- Alprazolam ✖
- Lorazepam
- Diazepam
- Chlordiazepoxide
- Nitrazepam

Your answer is incorrect.

Diazepam has the most rapid onset of action among orally administered benzodiazepines.

The correct answer is: Diazepam

Question 93

Correct

Mark 1.00 out of 1.00

You are treating an Afro-Caribbean man with schizophrenia using clozapine. What is the biggest risk factor for developing reduced white cell count?

Select one:

- Male sex
- Ethnicity ✔
- Dose-related
- Duration of treatment
- Treatment setting

Your answer is correct.

There is a 77% increase in risk for Afro-Caribbean people to develop clozapine-induced neutropenia. Other risk factors include young age and having a low baseline white cell count.

The correct answer is: Ethnicity

Question 94

Correct

Mark 1.00 out of 1.00

Mrs. Gertrude Sessions is a 34-year-old woman with a history of epilepsy with generalised seizures. She experiences auras preceding these seizures that are characterised by a profound sense of familiarity with her surroundings, even if she is in an entirely new environment. Seizure activity in which of the following areas of the brains is most likely to account for these symptoms?

Select one:

- Temporal lobe ✔
- Frontal lobe
- Parietal lobe
- Occipital lobe
- Cerebellum

Your answer is correct.

Seizures originating in the medial temporal lobe are most likely to lead to a sense of déjà vu.

The correct answer is: Temporal lobe

Question 95

Correct

Mark 1.00 out of 1.00

A 62-year-old woman with a history of hypertension and hypercholesterolemia presented with sudden onset of weakness of the right side. On examination, she had mild aphasia, with notable difficulties in word finding and naming objects shown to her. Which of the following brain structures is crucial for the function of object naming?

Select one:

- Parietal lobe ✔
- Amygdala
- Broca's area
- Hippocampus
- Wernicke's area

Your answer is correct.

"Two gyri in the **parietal lobe** are important to locate and become familiar with in regard to language. The first is the supramarginal gyrus (area 40), which curves around the posterior end of the lateral sylvian fissure. The second, the angular gyrus (area 39), lies directly posterior to the supramarginal gyrus. It curves around the end of a prominent sulcus in the temporal lobe, the superior temporal sulcus (see Fig. 2-5). Damage in the area of the **angular gyrus** in the dominant left hemisphere may cause word-finding problems (anomia), reading and writing deficits (alexia with agraphia), as well as left-right disorientation, finger agnosia (inability to identify the fingers), and difficulty with arithmetic (acalculia)." Excerpt from Wanda G.W. Organization of the Nervous System I, In Neurology for the Speech-Language Pathologist (Sixth Edition), Mosby, 2017, p 13-43.

The correct answer is: Parietal lobe

Question 96

Correct

Mark 1.00 out of 1.00

Javier Delgado is a 55-year-old man who, for the past 6 months, has experienced muscle twitches and spasms, ataxia, slurred speech and paraesthesia in different areas of the body. According to the neurologists, one of the differential diagnoses is CJD. He undergoes MRI investigation. If he does indeed have CJD, the following areas of the brain are likely to be affected with the exception of:

Select one:

- Pineal gland ✔
- Thalamus
- Cortex
- White matter
- Basal ganglia

Your answer is correct.

CJD is characterised by bilateral basal ganglia hyper-intensities (Pulvinar sign) on T2W and DWI MRI.

The correct answer is: Pineal gland

Question 97

Correct

Mark 1.00 out of 1.00

Wilfred Wren is an executive director of a financial company. As part of his work he has to fly to numerous countries, spanning several time zones. He finds that this is having a significantly disruptive effect on his sleep-wake cycles, and that it is taking him a long time to readjust after a long—haul flight involving several time zone changes. He consults you over the psychological impact of his sleep deprivation. He's been reading up on circadian rhythms and asks you which part of the brain is involved in the regulation of circadian rhythms. The correct answer is:

Select one:

- The suprachiasmatic nucleus ✔
- The primary somatosensory cortex
- The rhinencephalon
- The pituitary gland
- The cuneate nucleus

Your answer is correct.

The **suprachiasmatic nucleus** is a region of the brain in the hypothalamus, situated directly above the optic chiasm, responsible for controlling circadian rhythms, by communication with many other regions of the brain and the use of various peptides and neurotransmitters.

The correct answer is: The suprachiasmatic nucleus

Question 98

Correct

Mark 1.00 out of 1.00

MAO-A metabolizes

Select one:

- Noradrenaline and 5-HT only
- Glutamate and GABA
- Dopamine, Tyramine and phenyl ethylamine
- Noradrenaline, 5-HT and tyramine ✔
- Dopamine and Tyramine only

Your answer is correct.

MAO-A metabolizes noradrenaline, 5-HT and tyramine.

MAO-B metabolizes dopamine, tyramine and phenethylamine.

The correct answer is: Noradrenaline, 5-HT and tyramine

Question 99

Incorrect

Mark 0.00 out of 1.00

Which of the following enzymes in the monoamine synthetic pathway is largely restricted to adrenal medulla?

Select one:

- Tyrosine-3-hydroxylase
- Phenylethanolamine-N-methyltransferase
- L-Aromatic amino acid decarboxylase ✖
- Dopamine-beta-hydroxylase
- Phenylalanine-4-hydroxylase

Your answer is incorrect.

The occurrence of phenylethanol-amine-N- methyltransferase is largely restricted to the adrenal medulla, but with detectable levels in association with epinephrine neurons in the brain. Inhibition of enzyme activity decreases epinephrine biosynthesis. There is, however, a less specific N-methyl-transferase present in many tissues.

The correct answer is: Phenylethanolamine-N-methyltransferase

Question 100

Correct

Mark 1.00 out of 1.00

Which of the following is correct with respect to the use of LSD?

Select one:

- It is often injected
- Frequent use commonly causes tolerance
- Onset of action is delayed for 2 hours in most subjects
- Physiological withdrawal is not seen ✔
- It is available as a prescribed drug

Your answer is correct.

LSD is primarily used via oral administration but can be inhaled, injected, and transdermally applied. The onset of effects is rapid following intravenous administration (10 minutes). Following oral ingestion, the onset of the first effects are experienced in 20-30 minutes. Frequent, repeated doses of LSD are unusual, and therefore tolerance is not commonly seen. Tolerance does develop to the behavioral effects after 3-4 daily doses, but no withdrawal syndrome has been described.

The correct answer is: Physiological withdrawal is not seen

Question 101

Correct

Mark 1.00 out of 1.00

Which of the following findings is seen in patients who have attempted to commit suicide?

Select one:

- Reduced levels of prolactin
- Reduced 5-HIAA ✔
- Reduced glutamate levels
- Reduced levels of TSH
- Reduced choline levels

Your answer is correct.

Reduced homovanillic acid/5-hydroxyindolacetic acid (HVA/5-HIAA) ratios in cerebrospinal fluid (CSF) in depressed suicide attempters have been reported. The levels of norepinephrine (NE), dopamine (DA) serotonin (5-HT) and its metabolite 5-hydroxyindole acetic acid (5-HIAA) are reduced in those attempting suicide compared to healthy controls. A large effect size drop in 5-HIAA suggested that CSF 5-HIAA level may be used as an early biochemical indicator of a suicidal attempt.

(Retrieved from http://www.ncbi.nlm.nih.gov/pubmed/17453696)

The correct answer is: Reduced 5-HIAA

Question 102

Incorrect

Mark 0.00 out of 1.00

If a patient with terminal cancer related pain is administered morphine for analgesia, which of the following mechanism could explain the action?

Select one:

- Release of substance P ✖
- Release of histamine
- Release of serotonin
- Release of glycine
- Release of glutamate

Your answer is incorrect.

It is believed that μ-receptor agonist morphine causes the release of histamine in neurons leading to H2 receptor activation and subsequent relief of pain.

The correct answer is: Release of histamine

Question 103

Correct

Mark 1.00 out of 1.00

Synthesis and storage of noradrenaline can be prevented by which of the following?

Select one:

- Guanethidine
- Tyramine
- Tryptophan
- Tyrosine
- Reserpine ✔

Your answer is correct.

Reserpine prevents the synthesis and storage of norepinephrine in sympathetic nerve terminals.

The correct answer is: Reserpine

Question 104

Correct

Mark 1.00 out of 1.00

Impulsive aggression is related to a disruption in which of the following neurotransmitter systems?

Select one:

- Glutamate
- Norepinephrine
- Dopamine
- Acetylcholine
- Serotonin ✔

Your answer is correct.

One of the most consistent findings in all of psychobiology is the link between **low serotonin** in the brain and impulsive, explosive acts of violence.

(Linnoila and Virkkunen, 1992)

The correct answer is: Serotonin

Question 105

Correct

Mark 1.00 out of 1.00

Which of the following is a peptide neurotransmitter?

Select one:

- Glutamate
- Serotonin
- Dopamine
- Substance P ✔
- Acetylcholine

Your answer is correct.

Substance P is a neuroactive peptide synthesized in the cell body on the rough endoplasmic reticulum.

The correct answer is: Substance P

Question 106

Correct

Mark 1.00 out of 1.00

In a patient with panic attacks secondary to phaeochromocytoma, which of the following metabolites indicate excess of noradrenaline?

Select one:

- Homovanilic acid
- Aspartic acid
- mCPP (m-chlorophenylpiperazine)
- Vanillylmandelic acid ✔
- 5-Hydroxyindoleacetic acid

Your answer is correct.

Vanillylmandelic acid - also called VMA, is the major metabolite used to detect phaeochromocytoma. It indicates noradrenaline excess.

mCPP is a metabolite of trazodone.

5-HIAA is a serotonin metabolite.

The correct answer is: Vanillylmandelic acid

Question 107

Correct

Mark 1.00 out of 1.00

In CJD, the amyloid plaques are prominent in which part of the brain?

Select one:

- Cerebellum ✔
- Raphe nuclei
- Corpus callosum
- Occipital cortex
- Hippocampus

Your answer is correct.

CJD is a type of multifocal dementia involving the cerebellar structures, where amyloid plaques are prominent.

The correct answer is: Cerebellum

Question 108

Incorrect

Mark 0.00 out of 1.00

Normal adults enter deep sleep from wakefulness before entering REM sleep. But infants enter REM sleep directly from wakefulness. When does the adult pattern of sleep develop in a child?

Select one:

- 12 months ✖
- 3 years
- 18 months
- 5 years
- 3 months

Your answer is incorrect.

Newborns sleep for 16 - 18 hours daily though the longest sleep episode lasts only 2.5 to 4 hours. Newborns have three types of sleep: quiet sleep (similar to NREM), active sleep (analogous to REM), and indeterminate sleep. Sleep onset occurs through REM, unlike adults where it occurs through NREM. In infants, each sleep episode is made of only one or two cycles (instead of 4 or more in adults). This infantile sleep architecture disappears when circadian rhythms are fully entrained.

Circadian rhythm development in the first 3 months includes:

- emergence of the 24-hour core body temperature cycle (1 month of age)
- progression of nocturnal sleeping (2 months of age)
- cycling of melatonin and cortisol hormones in a circadian rhythm and **sleep onset begins with NREM (3 months of age)**

http://www.ncbi.nlm.nih.gov/books/NBK19956/

The correct answer is: 3 months

Question 109

Correct

Mark 1.00 out of 1.00

A 40-year-old man presents with gradually developing darkening of the skin most visible on scars, skin folds and pressure points. Following a trivial driving accident, he notices extreme fatigue, muscle weakness and weight loss. The most commonly used definitive test for this condition is:

Select one:

- Skin Biopsy
- ELISA serology
- Prolactin Test
- ACTH stimulation test ✔
- Fasting glucose

Your answer is correct.

This patient has **Addison's disease** which, due to insidious onset, often goes unnoticed until something else brings the symptoms to notice. The **ACTH stimulation test** is the most commonly used test for diagnosing adrenal insufficiency. In this test, blood cortisol, urine cortisol, or both are measured before and after a synthetic form of ACTH is given by injection. The normal response after an ACTH injection is a rise in blood and urine cortisol levels. People with Addison's disease or long-standing secondary adrenal insufficiency have little or no increase in cortisol levels.

The correct answer is: ACTH stimulation test

Question 110

Correct

Mark 1.00 out of 1.00

During relaxed periods of wakefulness, a normal individual's EEG will display which of the following rhythms?

Select one:

- Delta rhythm
- Gamma rhythm
- Alpha rhythm ✔
- Beta rhythm
- Theta rhythm

Your answer is correct.

During states of quiet wakefulness or drowsiness, the EEG pattern becomes slower (8 to 13 Hz with an average amplitude of 50 V) than what is seen during an alert state. This constitutes the alpha rhythm.

The correct answer is: Alpha rhythm

Question 111

Correct

Mark 1.00 out of 1.00

Raclopride is a compound used in imaging brain receptors. Which of the following receptors are blocked by raclopride?

Select one:

- D2 receptors ✔
- D1 autoreceptors
- 5-HT2C receptors
- 5-HT3 receptors
- 5-HT2A receptors

Your answer is correct.

Raclopride is a D2 antagonist that can be radiolabelled with the carbon-11 radioisotope for use in positron emission tomography (PET) scanning to assess the distribution of D2 receptors and the degree of endogenous dopamine binding to this receptor.

The correct answer is: D2 receptors

Question 112

Correct

Mark 1.00 out of 1.00

What is a risk factor for antidepressant-induced SIADH (syndrome of inappropriate ADH secretion)?

Select one:

- Low body weight ✔
- Male sex
- High initial sodium level
- Cold weather
- Tuberculosis

Your answer is correct.

Risk factors for antidepressant-induced SIADH include: Old age, low BMI, concomitant use of diuretics and female gender.

The correct answer is: Low body weight

Question 113

Correct

Mark 1.00 out of 1.00

Rosie is a 25-year-old woman brought into A&E by her boyfriend who came home to find her very confused. He notes that she is on some medication but doesn't know what it is. Her vital signs include pyrexia and a labile blood pressure, and on examination clonus in ankles. The most likely diagnosis is:

Select one:

- Serotonin syndrome ✔
- Neuroleptic malignant syndrome
- Meningitis
- Encephalitis
- Malignant hyperthermia

Your answer is correct.

Serotonin syndrome is the excess of serotonin resulting in a constellation of symptoms including confusion, agitation, lability in blood pressure and temperature, tremor and clonus. To differentiate between neuroleptic malignant syndrome and serotonin syndrome - onset for serotonin syndrome is much shorter (usually < 12hours), it is associated with hyperreflexia and clonus (NMS is associated with hyporeflexia), dilated pupils and also hyperactive bowel sounds.

(Ref: http://www.medsafe.govt.nz/profs/PUArticles/Dec2012Neuroleptic.htm)

The correct answer is: Serotonin syndrome

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